

Exercise 1. Solve Exercise 8 on page 168.

Exercise 2. Let X be a Banach space and let $T : X \rightarrow X$ be a bounded linear map that commutes with every other bounded linear map $M : X \rightarrow X$. Show that T must be a multiple of the identity map.

Exercise 3. Solve Exercise 11 on page 171.

Exercise 4. Solve Exercise 12 on page 171.

Exercise 5. Recall that the bounded linear map $M : X \rightarrow Y$ (X, Y Banach spaces) is bounded below, i.e.

$$\|Mx\| \geq c\|x\|,$$

for some fixed $c > 0$ and all $x \in X$. Prove that the following are equivalent:

- (i) M is bounded below,
- (ii) M is injective and its range is closed,
- (iii) M' is surjective.

Exercise 6. Let $K : [0, 1] \times \mathbb{R}^n \rightarrow \mathbb{C}$ have the properties:

- (i) $K(x, \cdot) \in L^2(\mathbb{R}^n)$ for all $x \in [0, 1]$,
- (ii) For every $f \in L^2(\mathbb{R}^n)$ the function

$$x \rightarrow \int_{\mathbb{R}^n} K(x, y) f(y) dy$$

is continuous on $[0, 1]$.

Prove that the integral operator $\mathbf{K}$ defined by

$$\mathbf{K}f(x) = \int_{\mathbb{R}^n} K(x, y) f(y) dy$$

is bounded from $L^2(\mathbb{R}^n)$ to $C([0, 1])$.

Exercise 7. Let $(X, \|\cdot\|)$ be a Banach space such that

- 1) $X \subset C([0, 1])$,
- 2) For every $r \in \mathbb{Q} \cap [0, 1]$ $f \rightarrow f(r)$ defines a bounded linear functional on X .

Prove that there exists $C > 0$ such that for all $f \in X$ we have

$$\sup_{x \in [0, 1]} |f(x)| \leq C\|f\|.$$

Theorem 0.1. *text*

Proof.

